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SUMMARY

I work on high-assurance systems security: Trusted Execution Environments, cryptographic attestation and root-of-trust, and hypervisor/OS isolation, paired with the adversarial side (side-channel attacks, exploit development, and fuzzing) to test whether those guarantees hold. Throughout my career, I have tackled security problems across the full hardware-software stack, driven by a desire to turn complex research problems into production-ready solutions.

WORK EXPERIENCE (10+ yrs research, 2+ yrs industry)

Senior Security Engineer: NVIDIA Corporation, Santa Clara, CA **Jul 2024 – Current**

- Implemented end-to-end Vera CCA and Baremetal Secure AI attestation, from firmware tokens the attester retrieves.
- Implemented CoRIM generation from golden reference values, released to customers through FLM and rim-service.
- Implemented verifier tools that check the evidence against the released CoRIM.
- Implemented Live Firmware Activation (LFA) for the RMM, building the stack across RMM, ATF, and Vera IROT.
- Protected the NVFlare model so the CVM decrypts its disk and loads it only after initramfs releases the key.

Postdoc Researcher: Georgia Institute of Technology, Atlanta, GA **Jan 2022 – Jul 2024**

- Designed PORTAL, bringing DMA devices inside the trust boundary of Arm CCA Realm VMs [1].
- Built autofz to choose and schedule fuzzers from live coverage during a campaign [4].
- Researched potential threats on Intel TDX and Arm CCA, where the hypervisor stays outside the TCB.

Postdoc Researcher: Korea Advanced Institute of Science and Technology, Korea **Mar 2021 – Dec 2021**

- Cache side-channel research on Intel and Apple M1, published as PRIME+RETOUCH [3].

Research Intern: Georgia Institute of Technology, Atlanta, GA **Nov 2019 – May 2020**

- Prototyped enclave-visible cache and TLB notifications, published as SENSE [2].

Research Intern: Microsoft Research, Redmond, WA **May 2016 – Aug 2016**

- Developed a blind ROP attack on Intel SGX enclaves whose binaries stay hidden from the attacker [6].
- Developed a secure CDN that lets enterprises offload sensitive user data to an untrusted third-party cloud.

TECHNICAL SKILLS

TEEs: Intel SGX, Intel TDX, Arm CCA, Arm TrustZone
Systems: Linux kernel, KVM, QEMU, gem5
Attestation: EAT, CoRIM/CoMID, DICE/DPE
Devices: SPDM, TDISP, SMMU/IOMMU, PCIe
Firmware: ATF, TF-RMM, Caliptra DPE, PSC (NVIDIA IROT)
Languages: C, C++, Python, Rust, ARM and x86 assembly, SystemVerilog, Chisel
Architecture: CPU cache/CPU pipeline, TLB/DRAM

PUBLICATION & PATENT (869 citations)

Selected TEE and platform-security work. Venues: ***S&P, USENIX Security, NDSS, TDSC****.*

[1] **PORTAL: Fast and Secure Device Access with Arm CCA for Modern Arm Mobile SoC** | Paper **SP 2025**

- Native Arm CCA encrypts data exchanged with devices, imposing performance and power overhead on mobile SoCs.
- Built PORTAL: GPT and SMMU stage-2 controls isolate plaintext shared memory to an assigned Realm and device.
- **Keywords:** Arm CCA, GPT, SMMU, DMA isolation.

[2] **SENSE: Enhancing Microarchitectural Awareness via Subscription-Based Notification** | Paper **NDSS 2024**

- An attacker can see cache and TLB events, but the enclave cannot trust the OS to report them.
- Added a gem5 subscription ISA that delivers cache and TLB evictions to the enclave at 1.2% overhead.
- **Keywords:** Intel SGX, cache side channel, TLB, gem5.

[3] **PRIME+RETOUCH: When Cache is Locked and Leaked** | Paper **arXiv 2024**

- Cache-locking defenses assume a secret leaks only when the protected line is evicted.
- Showed a Tree-PLRU leak on Intel and Apple M1 while the line stays resident; Intel acknowledged it.
- **Keywords:** cache side channel, Tree-PLRU, Intel TSX, Apple M1.

[4] **autofz: Automated Fuzzer Composition at Runtime** | Paper | Hacker News **USENIX 2023**

- No single fuzzer wins on every target, and the best choice changes during a campaign.
- A coordinator over 12 fuzzers reallocates CPU from coverage and finds 152% more bugs on UNIFUZZ and FTS.
- **Keywords:** fuzzing, runtime scheduling, coverage.

[5] **SGX-Bomb: Locking Down the Processor via Rowhammer Attack** | Paper | Hacker News | Demo **SysTEX 2017**
- SGX locks the processor on an integrity failure, assuming software cannot flip DRAM bits.
- Rowhammer inside an enclave flips memory-encryption metadata and locks the machine in 283 seconds.
- **Keywords:** Intel SGX, Rowhammer, DRAM integrity, denial of service.

[6] **Hacking in Darkness: Return-oriented Programming against Secure Enclaves** | Paper | Talk **USENIX 2017**
- SGX hides the enclave binary, so a memory-corruption bug has no visible gadgets to reuse.
- An AEX fault oracle rebuilds a gadget map, then exfiltrates keys and hijacks remote attestation.
- **Keywords:** Intel SGX, ROP, remote attestation.

[7] **PrivateZone: Providing a Private Execution Environment using ARM TrustZone** | Paper **TDSC 2016**
- TrustZone's Secure World is closed to third-party code and reserved for OEM services.
- Stage-2 page tables in Monitor mode keep unmodified processes out of the Secure World, at 3% overhead.
- **Keywords:** Arm TrustZone, stage-2 paging, TEE.

[8] **OpenSGX: An Open Platform for SGX Research** | Paper | Wikipedia: SGX **NDSS 2015**
- No public way existed to build or study SGX enclaves before silicon and Intel's closed emulator.
- Built a QEMU SGX platform with 308 GitHub stars.
- **Keywords:** Intel SGX, QEMU, software emulation.

[9] **Apparatus and method for open and private IoT gateway using Intel SGX** | Patent **KR102162018B1**
- An IoT gateway may convert sensitive traffic in a domain the operator does not fully trust.
- Intel SGX holds protocol conversion, encrypted transfer, and key management.
- **Keywords:** Intel SGX, IoT gateway, key management.

SOFTWARE ARTIFACTS

[1]	**Autofz:**	https://github.com/sslab-gatech/autofz	⑂11	★85
[2]	**OpenSGX:**	https://github.com/sslab-gatech/opensgx	⑂82	★310
[3]	**SGXBomb:**	https://github.com/sslab-gatech/sgx-bomb	⑂4	★18

PROFESSIONAL ACTIVITIES

Journal/Conference Reviewer

IEEE TDSC	*IF 7.3, 2024*	Computers & Security	*IF 5.7, 2024*
JISA	*IF 5.6, 2024*	IEEE Access	*IF 3.9, 2024*
ACM TOIT	*IF 3.9, 2024*	JCN	*IF 3.6, 2024*
IEEE S&P	*2016–2018, 2020–2022*	USENIX Security	*2015, 2021–2024*
ACM CCS	*2015, 2017–2019, 2023*	NDSS	*2019–2021*
SOSP	*2021*	WWW	*2018*
ASIACCS	*2015–2016, 2018*	IEEE MASS	*2024*
IEEE ICPADS	*2024*		

INVITED TALKS

[1] **A Novel Cache Side Channel Attack without Attacker Eviction**
- Presented at Intel Side Channel Academic Program (SCAP), Virtual, Sep 2020

[2] **Hacking in Darkness: Return-oriented Programming against Secure Enclaves**
- Presented at the 26th USENIX Security | Presentation Link, Vancouver, BC, Canada, Aug 2017
- Presented at KimchiCon | Presentation Link, Daejeon, South Korea, Aug 2017

[3] **Tutorial: Trusted Execution Environment: TrustZone, ITP and SGX**
- Korea Institute of Information Security and Cryptology, Seoul, South Korea, Apr 2015

EDUCATION

Institution	Degree	Field of Study	Dates	GPA
Korea Advanced Institute of Science and Technology	*Ph.D.*	Information Security	Aug 2015 – Feb 2021	3.95
Korea Advanced Institute of Science and Technology	*M.S.*	Information Security	Sep 2013 – Aug 2015	4.1
Handong Global University	*B.S.*	Computer Science	Mar 2010 – Aug 2013	4.2

THESIS

Unpremeditated Usage of Benign System Components Devastates Security — Ph.D. Thesis.

Detecting Emulated Environment: Exploiting Behavioral Discrepancies in QEMU — M.S. Thesis.